# PureRoute v2 Hardware Playbook

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Build one leak-free bench rig: a water loop with a filter cartridge that can clog, a sensing cup for test samples, and an Arduino Uno that reads flow, turbidity, conductivity and temperature and drives the pump and four state LEDs. A laptop connected by USB runs the dashboard.

## What the rig must do

1. Tap water in the sensing cup → green (UF+UV).
2. Salty water → blue (RO).
3. Muddy water → yellow blinking (FLUSH).
4. Clogging the sealed cartridge drops the flow, the manometer tube rises, and the state goes to yellow REPLACE.
5. Lifting the pump out of the water → red, and the pump stops by itself.

LEDs stand in for the purifier's valves.

## Parts

**Already have:**

- Arduino Uno with its USB A-to-B cable
- A laptop for the dashboard (a Raspberry Pi 4 also works, but isn't needed)
- DS18B20 waterproof temperature probe (CentIoT, 1 m)
- Turbidity sensor module (washing-machine type, with small board)
- Mini 3–6 V submersible pump with 1 m clear tube
- YF-S401 flow sensor (0.3–6 L/min)

**Buy tonight (local electronics and hardware shops):**

| Item | Qty | Why |
|---|---|---|
| D4184 or LR7843 MOSFET module | 1 | Switches the pump from the Uno |
| 1N5819 or 1N4007 diode | 2 | Flyback protection across the pump |
| 5 V 2 A adapter (old phone charger) | 1 | Separate pump supply, so the Uno doesn't reset |

| Item | Qty | Why |
|---|---|---|
| Capacitors: 470–1000 µF | 1 | Across the pump supply |
| Capacitors: 100 nF | 3 | Pump supply, turbidity signal, spare |
| LEDs: green, blue, yellow, red | 1 each + spares | Route and state display |
| Resistors | See list below | Pull-up, pull-down, LEDs, probe |
| Stainless steel (SS304) M3 or M4 screws with nuts | 4 | Conductivity probe electrodes (2 + spares). Not copper or aluminium. |
| Wide-mouth plastic jar with a tight screw lid | 2 | Sealed filter cartridge (fresh + clogged) |
| Clear cup or cut bottle | 2 | Sensing cup and reservoir |
| Extra clear tube about 30 cm, tube clamps or zip ties | — | Manometer and secure joints |
| Hot glue or epoxy, PTFE tape, black insulation tape | — | Sealing and light shielding |
| Sponge, cotton cloth, optional activated carbon | — | Filter media |
| Table salt, fine soil or chalk, milk | — | Test samples |
| Breadboard and jumper wires, or perfboard and wire | — | Connections |

**Resistors:**

- 1 kΩ × 2 (probe series, spare)
- 4.7 kΩ × 1 (DS18B20 pull-up)
- 10 kΩ × 3 (pump pin pull-down, optional probe low range, spare)
- 220–330 Ω × 4 (LEDs)

**Borrow from the lab:**

- Multimeter: essential, to check voltages before they touch the ESP32
- Handheld TDS/EC pen as the reference meter
- 0.01 g scale for weighing salt
- Measuring jug
- Stopwatch (phone)

Keep every bill.

## Bench layout

The rig has two parts on one rigid board (plywood or thick cardboard, about 40 × 30 cm), so nothing moves during filming.

```
FILTER LOOP (clean tap water only)

  reservoir --> pump --> flow sensor --> tee --> sealed cartridge jar -->
back to reservoir
   (bucket)   (submerged)  (YF-S401)      |       (sponge / cloth)
(outlet above waterline)
                                        |
                              vertical clear tube = manometer
                              (water rises as the filter clogs)

SENSING CUP (separate, filled by hand with each sample)

  card lid holding: turbidity sensor tip | conductivity probe | DS18B20
  probes 2-3 cm apart, cup wrapped in black tape against room light

ELECTRONICS (raised, at least 15 cm from any water joint, tray underneath)

  Arduino Uno + MOSFET module + 4 LEDs (green UF, blue RO, yellow
flush/replace, red fault)
  Laptop running the dashboard, connected to the Uno by USB
```

**Why two parts:**

- The loop shows the filter clogging (flow and pressure).
- The sensing cup checks the incoming water (salt, mud, temperature).
- Keep salt and mud out of the pump loop: soil damages the small pump, and salt corrodes it.
- Clog the cartridge only with cloth layers or a spoon of fine soil rubbed into the cloth.

## Power and pin map

| Function | Uno pin | Wiring |
|---|---|---|
| DS18B20 data | D4 | 4.7 kΩ pull-up from data to 5 V; red → 5 V, black → GND |

| Function | Uno pin | Wiring |
|---|---|---|
| Flow sensor signal | D2 (interrupt) | Red → 5 V, black → GND, yellow → D2 directly. No divider needed. |
| Turbidity analog out | A0 | Module on 5 V; AOUT → A0 directly, 100 nF from A0 to GND. Leave its digital output unconnected. |
| Conductivity sense | A1 | Joins electrode 1 and the 1 kΩ resistor |
| Conductivity drive A | D7 | Through 1 kΩ to electrode 1 |
| Conductivity drive B | D8 | Directly to electrode 2 |
| Conductivity low range (optional) | D9 | Through 10 kΩ to the sense node, for very pure water |
| Pump control | D5 | To the MOSFET module's IN/PWM, plus a 10 kΩ pull-down from D5 to GND |
| LEDs | D10 green, D11 blue, D12 yellow, D13 red | Each through its own 220–330 Ω to GND |
| Dashboard link | USB | Uno USB-B to the laptop; this also powers the Uno |

**Rules:**

- The Uno runs on 5 V, so 5 V sensor outputs can connect directly. Still check each signal with the multimeter: nothing should exceed 5 V.
- Flow must be on D2 or D3, the only interrupt pins on the Uno.
- **The pump gets its own 5 V 2 A supply.** Running it from the Uno's 5 V pin causes resets.
- Join the pump supply ground and the Uno ground at one point.
- Fit the 470–1000 µF and 100 nF capacitors across the pump supply, near the MOSFET module.
- Fit the flyback diode across the pump terminals, band (cathode) to pump +. The D4184 module has none built in.
- The 10 kΩ pull-down keeps the pump off while the Uno is booting or resetting.
- Sensors draw under 100 mA in total, which the Uno's 5 V pin supplies from USB.

```
 +5V pump supply ──┬── 470-1000 µF ── GND
                   ├── MOSFET VIN+ ; VIN- ── GND
 D5 ──┬── MOSFET IN    MOSFET OUT+ ── pump + ─┐
      └─[10k]─ GND   MOSFET OUT- ── pump - ─┘   (diode across pump, band to
+)

 Flow:      red→5V, black→GND, yellow→D2
 Turbidity: VCC→5V, GND, AOUT→A0 (100 nF to GND)
 DS18B20:   red→5V, black→GND, yellow→D4, 4.7k from D4 to 5V

 Conductivity probe (2 stainless screws, 8-15 mm apart):
 D7 ──[1k]──┬── electrode 1        D8 ──────── electrode 2
 D9 ──[10k]─┤ (optional)
            └── A1 (sense)

 Uno USB ──── laptop USB (data + Uno power)
```

# Build steps

## Conductivity probe

1. Drill two holes 8–15 mm apart in a rigid plastic piece (a PVC pipe cap or thick lid).
2. Push the two stainless screws through and lock them with nuts, so the spacing can never change.
3. Mark the immersion depth on the holder. Every reading must be taken at that same depth.
4. Solder or crimp wires to the screw heads and seal those joints with hot glue or epoxy, above the waterline.
5. Clean the screw tips with vinegar, then rinse with distilled or RO water.

Spacing and depth set the probe's calibration. If they shift, recalibrate.

## Sealed cartridge jar (the most important build)

1. Make two holes in the jar lid, sized to the tube's outer diameter.
2. Inlet tube goes through the lid down to the jar bottom; outlet stub ends just under the lid. Water flows up through the media.
3. Pack from the bottom: sponge disc, folded cotton cloth pad, optional carbon in a cloth pouch, second sponge disc pressed under the lid. Pack firmly so water can't go around the media.

4. Seal both tube holes with hot glue on both sides of the lid. Add a rubber gasket (cut from an old bicycle tube) or silicone under the lid.
5. Make two jars: **Fresh**, and **Clogged** (4 extra cloth layers plus a spoon of fine soil rubbed into the cloth). For the ramp test, add cloth layers to the fresh jar step by step.

**If the jar isn't sealed, flow will never drop when it clogs.** Water just escapes, and the main demo fails.

### Sensing cup

- Clear cup wrapped in black tape. Room light disturbs the turbidity reading.
- A card or plastic lid with three holes holds the turbidity tip, the conductivity probe and the DS18B20, at least 2–3 cm apart.
- Only the clear optical tip of the turbidity sensor goes in the water, never its circuit board.

### Manometer (cheap pressure proxy)

- Put a T-joint in the tube just before the cartridge.
- Run 30 cm of clear tube straight up from it, open at the top, taped to a ruler.
- As the filter clogs, the water column rises. Film it: it's visual proof of back-pressure, and it shows where a pressure sensor would go in the production version.

### Loop plumbing

- Pump submerged at the bottom of the reservoir, with its cable joint above the water.
- Flow sensor horizontal, with the arrow in the flow direction and 10 cm of straight tube before it.
- Short tubes, no kinks, a clamp or zip tie on every joint, and PTFE tape on threaded ends.
- Return tube ends above the reservoir waterline.

## Bring-up: test each part alone

Test each part on its own before combining anything. Each step needs a tiny test sketch from the software team, or just a serial print.

| Step | Test | Pass when |
|---|---|---|
| 1 | Leak test: pump on its adapter alone, electronics disconnected, paper towel under every joint, 10 min | No drips; water returns to reservoir |
| 2 | Uno connected to the laptop; open the Arduino IDE Serial Monitor at 115200 baud | Text arrives from the Uno |
| 3 | DS18B20 prints temperature | Room temperature, steady; not −127 or 85 |
| 4 | Flow sensor: blow gently through it, then run the loop | Pulse count rises; roughly steady L/min with pump on |
| 5 | Turbidity: measure AOUT with the multimeter first | Clear water about 4.0–4.5 V, never above 5 V |
| 6 | Conductivity: probe in tap water, then salty water | Reading clearly changes; probe in air reads out-of-range |
| 7 | Pump from D5 through the MOSFET, on its own supply | Pump starts and stops; Uno does **not** reset; pump stays off while the Uno reboots |
| 8 | Fresh jar vs clogged jar | Flow clearly lower with clogged jar; manometer column higher |

**Step 8 is the biggest risk.** If the clogged jar reads zero flow, it's below the sensor's 0.3 L/min minimum. Use fewer cloth layers so the ramp shows a clear drop that stays measurable.

## Calibration

Record every calibration in a CSV with the date and room temperature, and send the constants to the software team.

### Flow constant K (15 minutes)

Listings disagree (5,880 vs 4,380 pulses per litre), so measure it yourself.

1. Run the loop into a measuring jug for exactly 60 s and note the pulse count and volume.
2. Repeat 3 times with the fresh jar and 3 times with the clogged jar.
3. K = pulses ÷ litres. Expect 4,000–6,000.

4. With the fresh jar, record the healthy flow Q0 (the software team saves it with the `F` command).

### Conductivity (45 minutes)

1. Weigh table salt into 1 L of RO or distilled water at 0, 0.25, 0.5, 1.0 and 2.0 g per litre.
2. For each: dip the probe to its mark, wait 30 s, take 5 readings, then rinse with RO water and blot dry.
3. Measure each solution with the borrowed TDS/EC pen and note the pen's ppm factor.
4. The software team fits a curve from probe reading to pen reading.
5. Only trust it when the room temperature at start and end match.

Expect about ±5–10% after calibration. Very pure water (RO output) will only read as "low".

### Turbidity (30 minutes)

1. In the black-taped cup with clear tap water, save the baseline (`B` command).
2. Add full-fat milk drop by drop to 250 mL: 0, 1, 2, 4, 8, 16 drops. Take 5 readings at each step.
3. Repeat with fine soil or chalk at about 0.05, 0.1, 0.5 and 1 g per litre, stirred.
4. Take one reading with room lights on and one with lights off, to show the shield works.

Report turbidity as a % index against clear water, **never as NTU**. These modules vary a lot between units.

## Real water samples

Test the rig on real local water as well as the salt and mud samples. Collect about 250 mL of each in clean labelled bottles: source, date, time.

| Sample | Where | Expected (to be confirmed by measuring) |
|---|---|---|
| RO output | Any RO purifier on campus or at home | Lowest conductivity |
| RO reject | Same purifier's waste line | Highest conductivity |
| Bottled water | Any brand | Low |
| Tap / municipal | Hostel or home tap | Low to medium |

| Sample | Where | Expected (to be confirmed by measuring) |
|---|---|---|
| Borewell / hand pump | If available nearby | Medium or higher |
| Rain or tanker | If available | Varies |

**How to measure:**

- Measure each in the sensing cup 3 times, plus once with the TDS pen.
- Log conductivity, turbidity index and temperature.
- An unexpected order is a finding, not a mistake. Report it as measured.
- Don't quote Haldwani TDS figures from the internet: no official local figure was found, so your own measurements are the data.

Never drink anything from the rig.

## Common problems and fixes

| Symptom | Likely cause | Fix |
|---|---|---|
| Uno resets when the pump starts | Pump on the Uno's 5 V | Separate pump supply, big capacitor, flyback diode |
| Pump runs briefly at power-up | No pull-down on D5 | Add 10 kΩ from D5 to GND |
| Flow reads 0 with pump on | Airlock, pump not primed, sensor backwards, wrong pin | Submerge pump fully, check the arrow, tilt tube to bleed air, use D2 |
| Flow never drops when clogged | Jar not sealed | Re-seal lid and tube holes; inlet to bottom, outlet at top |
| Flow reads double or half | Wrong K | Redo the jug calibration |
| Conductivity drifts up over minutes | Probe polarising, bubbles | Tap bubbles off, clean screws with vinegar, confirm the code uses both pulse directions |
| Conductivity jumps between dips | Depth changes | Use the depth mark or a fixed holder |

| Symptom | Likely cause | Fix |
|---|---|---|
| Turbidity changes when people move | Room light | Black tape, fixed lighting |
| Dashboard can't find the Uno | Wrong port or Serial Monitor open | Check the port in Device Manager (Windows) or `/dev/ttyACM*` (Mac/Linux); close the Serial Monitor before running the dashboard |
| DS18B20 shows −127 °C | Missing pull-up or wrong pin | 4.7 kΩ from data to 5 V |
| Pump hums, no water | Lift too high or intake blocked | Lower the outlet, clean the intake grille |
| Leak at flow sensor | Tube size mismatch | Clamp, PTFE tape, hot glue |

## Safety

- Only low voltage (5 V) near water. No mains wiring on the board; plug adapters in away from the water.
- Electronics raised and away from joints, with a tray underneath.
- Leak-test before connecting electronics.
- Label the rig "Test bench, not drinking water".
- Handle the multimeter carefully around wet parts; dry hands before touching the circuit.

## Build order

- [ ] Buy everything on the parts list; borrow the multimeter, TDS pen and scale
- [ ] Build the board base, both cartridge jars, the sensing cup and the conductivity probe
- [ ] Plumb the loop and run the 10-minute leak test
- [ ] Wire the pin map, checking every signal with the multimeter before it touches the Uno
- [ ] Bring-up steps 1–8, especially fresh vs clogged flow
- [ ] Add the manometer tube
- [ ] Calibrate flow, conductivity and turbidity; give the constants to the software team
- [ ] Collect the real water samples
- [ ] Run the clogging ramp, sample and fault tests with the software team
- [ ] Photograph the rig, jars, probe, wiring, each LED state and the manometer